SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	**Assessment:**	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO5 Statement:	*Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements.(BL5)*		
Student Name:	Kola Amrutha Sandhya	**Reg. No.:**	192524270
Section / Batch:	AIDS	**Date:**	27-08-26

Code of Conduct

Ikola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just "make it better."
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Analysis Report

CO5 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO5-AT2 Visualization Analysis Report problems, in which students critique a described visualization design choice, identify its specific flaw, and justify a well-reasoned improvement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., proportional ink, axis scale choice, overplotting, color scale design, accessibility).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	**25%**	Clearly explains why the design choice is flawed or effective, including its practical consequence for a reader — not just that it is 'good' or 'bad.'	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	**20%**	Proposes a specific, well-justified fix or recommendation that directly addresses the evaluated issue.	Proposes a workable fix, with only minor gaps in justification.	Proposes a fix, but it only partially addresses the issue or lacks justification.	Fails to propose a workable fix, or it does not address the evaluated issue.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	**15%**	The response is clearly structured, addressing critique and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	**10%**	Both the critique and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the critique or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the critique and the recommendation.

Scoring Guide

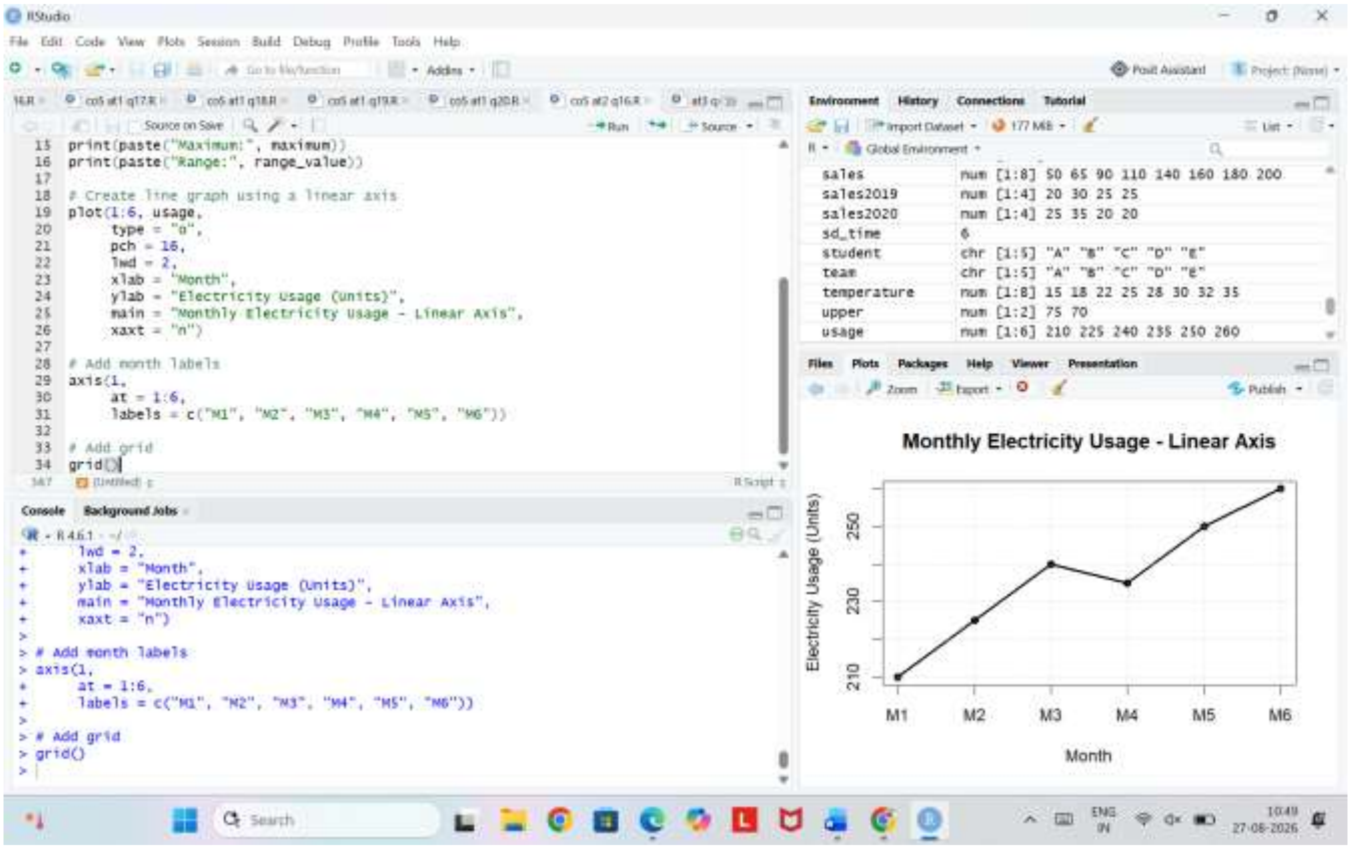
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 16

Data: Data: monthly electricity usage: 210, 225, 240, 235, 250, 260 units.

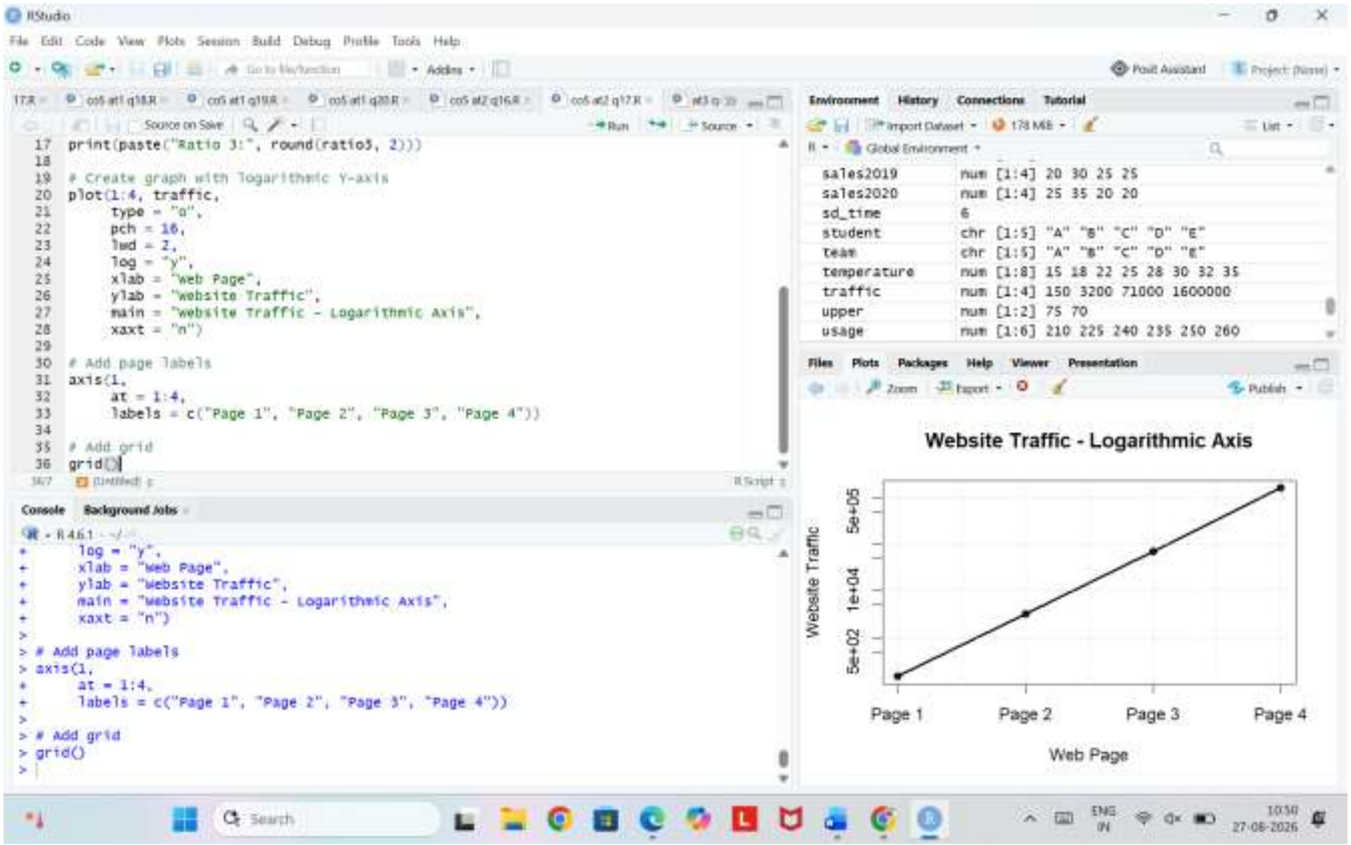
Task: Explain why a linear axis is the appropriate choice for this data, referencing its range.



Problem 17

Data: Website traffic across 4 pages: 150, 3200, 71000, 1600000.

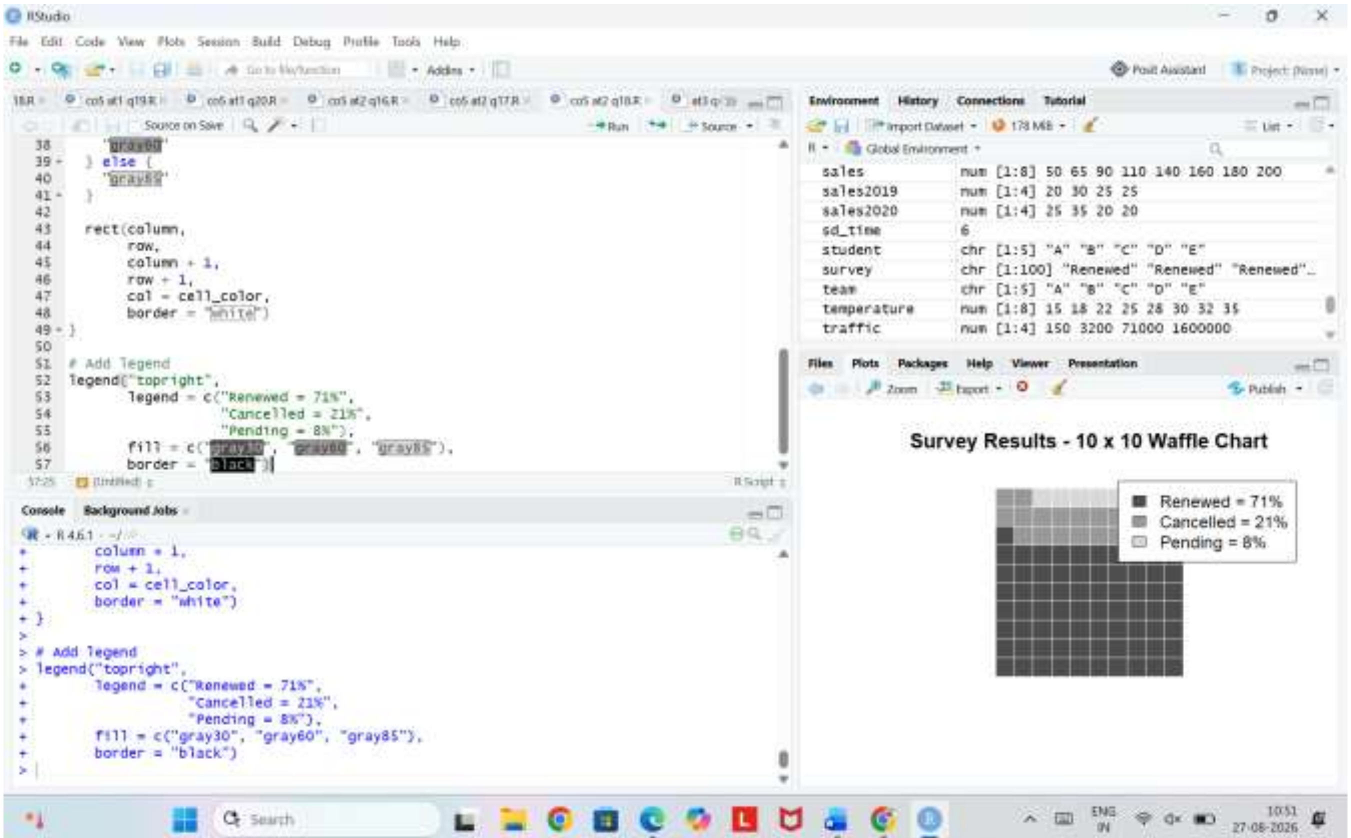
Task: Calculate the ratio between each consecutive value, and explain why a logarithmic axis is necessary here.



Problem 18

Data: Survey results: Renewed = 71%, Cancelled = 21%, Pending = 8%.

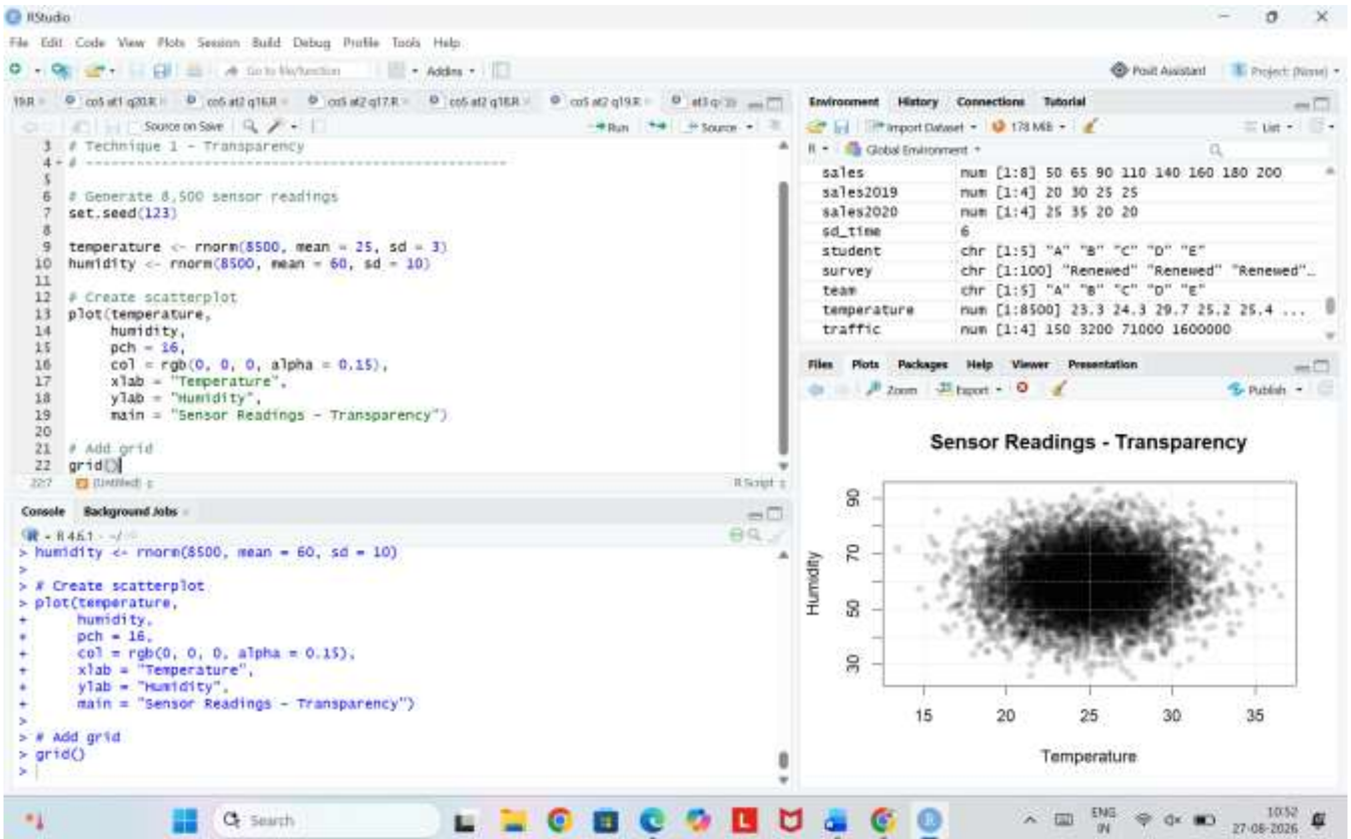
Task: Describe how you would build a 10×10 waffle chart for this data, stating exactly how many cells



Problem 19

Data: A scatterplot of 8,500 sensor readings plotted as temperature vs. humidity appears as a single solid blob.

Task: Suggest two different techniques to fix this overplotting problem, and briefly explain how each works.



Problem 20

Data: A categorical scatterplot of delivery times for 3 couriers (X, Y, Z) shows heavy point-stacking directly above each category label.

Task: Explain how combining jittering with partial transparency would improve this chart's readability.